

CCE

USN

--	--	--	--	--	--	--	--	--	--



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CC3214-1 – WIRELESS SENSOR NETWORKS
 (For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following best defines an active sensor?

A) Passively measures environmental conditions	B) Requires no energy input
C) Actively probes the environment	D) Measures without motion or energy
2. In a layered architecture for sensor networks, what is the primary role of the base station ?

A) Communication gateway to external networks	B) Direct communication with every sensor node
C) Central processing for all sensor data	D) Local computations for each sensor node
3. Which of the following is an example of a passive, omnidirectional sensor?

A) Camera	B) Radar
C) Thermometer	D) Sonar
4. Which of the following challenges are typically associated with WSNs?

A) Distributed execution, energy efficiency	B) Scalable power, structured topology
C) Energy constraints, unreliable communication	D) Fixed deployment, reliable sensors
5. In a Wireless Sensor Network (WSN), what is the primary role of the sinks?

A) receive data from source nodes	B) process data within the network
C) sense and collect data from the environment	D) control and power all sensor nodes
6. In RF communication, which component allows both transmitting and receiving signals?

A) Amplifier	B) Transceiver
C) Oscillator	D) Modulator
7. What is meant by "low access delay" in wireless MAC protocols?

A) The time between packet arrival and the first attempt to transmit it	B) The time taken to retransmit a packet after a collision
C) The time taken to detect a collision	D) The time between packet arrival and its successful delivery
8. The hidden-terminal problem occurs in which type of MAC protocol?

A) Frequency Division Multiple Access (FDMA)	B) Time Division Multiple Access (TDMA)
C) Code Division Multiple Access (CDMA)	D) Carrier Sense Multiple Access (CSMA)
9. Which of the following techniques is used to solve the hidden-terminal and exposed-terminal problems?

A) Time Division Multiple Access (TDMA)	B) Frequency Hopping Spread Spectrum
C) RTS/CTS handshake (Request to Send / Clear to Send)	D) Power control algorithms

- 
10. What varies in WSN deployments?
 - A) Network densities
 - B) Network protocols
 - C) Data transmission rates
 - D) Sensor sizes
 11. What causes changes in wireless connections?
 - A) Node mobility
 - B) Noise and interference
 - C) Battery depletion
 - D) Data aggregation
 12. How do sensors save energy when not needed?
 - A) Changing network topology
 - B) Data compression
 - C) Reducing transmission power
 - D) Duty cycle management
 13. What must routing protocols balance for efficiency?
 - A) Scalability, reliability, energy
 - B) Cost and complexity
 - C) Security and privacy
 - D) Data rates and latency
 14. What limits a packet's lifespan in flooding?
 - A) Node energy levels
 - B) Hop count or TTL
 - C) Bandwidth availability
 - D) Data packet priority
 15. What is a drawback of flooding in WSNs?
 - A) Packet duplication
 - B) Low data delivery rates
 - C) Complex route calculations
 - D) High latency
 16. How does gossiping avoid traffic implosion?
 - A) Packet filtering mechanisms
 - B) Using acknowledgments for packets
 - C) Predicting network congestion
 - D) Sending to one neighbor
 17. What functionality should a sensor operating system ideally provide?
 - A) High-level graphics support
 - B) Real-time support
 - C) Extensive file management
 - D) Complex user interfaces
 18. Which approach does TinyOS use to manage tasks?
 - A) Preemptive multithreading
 - B) Multicore processing
 - C) Event-driven model
 - D) Batch processing
 19. Why are traditional operating systems unsuitable for wireless sensor networks?
 - A) high resource demands.
 - B) too complex for users.
 - C) do not support networking.
 - D) extensive user input.
 20. How does an embedded system communicate with the outside world?
 - A) Input
 - B) Output
 - C) Memory
 - D) Peripherals

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Imagine you are designing a WSN for disaster management. What characteristic requirements must be prioritized to ensure effective operation? Explain with reasons.	8	L*2	1	1,2
	b) Draw a comparison table between MANET and WSNs, and explain which one is more suitable for: (i) Real-time military operations (ii) Environmental data collection	8	L2	1	1
2.	a) Explain and differentiate Hidden Terminal and Exposed Terminal Scenario.	8	L1	1	1
	b) Differentiate between collision handling in wired networks vs. wireless networks, and explain why collision detection protocols cannot be applied to wireless media.	8	L2	2	1,2

CC3214-1**SEE – December 2025**

- | | | | | | | |
|----|----|--|---|----|---|---|
| 3. | a) | Give the classes of MAC Protocol and explain them in detail. | 8 | L2 | 2 | 2 |
| | b) | What is TRAMA ? Explain its types in detail. | 8 | L1 | 2 | 2 |

Unit – II

- | | | | | | | |
|----|----|---|---|----|---|-----|
| 4. | a) | Enumerate the routing challenges and design issues in WSNs | 8 | L2 | 3 | 1,2 |
| | b) | With the network feature differentiate Data-Centric protocols and Data-Centric Routing. | 8 | L2 | 3 | 1,2 |
| 5. | a) | Explain the basic operations for sensor protocols for information via Negotiation (SPIN). | 8 | L1 | 3 | 2 |
| | b) | Explain Rumor Routing: its advantages & disadvantages. | 8 | L2 | 4 | 2 |
| 6. | a) | What is the LEACH Protocol? Explain the working with respect to WSNs. | 8 | L2 | 4 | 1 |
| | b) | "GAF protocol reduces energy consumption in WSNs without affecting routing fidelity." Explain this statement with the working mechanism of Geographic Adaptive Fidelity Protocol. | 8 | L3 | 4 | 1 |

Unit – III

- | | | | | | | |
|----|----|--|---|----|---|---|
| 7. | a) | Explain the MagnetOS and MANTIS Operating Systems for Wireless Sensor Network. | 8 | L2 | 5 | 1 |
| | b) | Explain the OSPM and EYES OS Operating Systems for Wireless Sensor Network. | 8 | L2 | 5 | 2 |
| 8. | a) | Discuss the Operating System Design Issues and Sensor operating systems (SOS) functions in WSNs. | 8 | L2 | 5 | 2 |
| | b) | Explain the PicOS and EMERALDS Operating Systems for Wireless Sensor Network. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
